P4 Firewall with DNS Deep Packet Inspection

Test Results Documentation

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March 25, 2026

Project Overview

Base: p4lang/tutorials firewall exercise (TCP Bloom filter)
Enhancement: P4DDPI paper (AlSabeh et al., NDSS MADWeb 2022)
Target: BMv2 software switch
Network: Mininet with 4 hosts, 4 switches

Test 1: Basic Connectivity (Ping Test)

Objective

Verify that the network is properly configured and packets can flow between internal and external hosts through the firewall.

Command

```
mininet> h1 ping h3 -c 3
```

Expected Result

- 3 packets transmitted
- 3 packets received
- 0% packet loss
- Round-trip time displayed

What This Tests

- Network topology is correct
- Routing tables are loaded
- ARP resolution works
- Firewall forwards legitimate traffic
- Basic IPv4 forwarding functions properly

Screenshot

```
mininet> h1 ping h3 -c 3
PING 10.0.3.3 (10.0.3.3) 56(84) bytes of data.
64 bytes from 10.0.3.3: icmp_seq=1 ttl=61 time=12.4 ms
64 bytes from 10.0.3.3: icmp_seq=2 ttl=61 time=5.96 ms
64 bytes from 10.0.3.3: icmp_seq=3 ttl=61 time=8.07 ms

--- 10.0.3.3 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 5.964/8.824/12.444/2.699 ms
```

Status: **PASS** - Network connectivity established successfully

Test 2: DNS Blocking (Malicious Domain)

Objective

Verify that the firewall blocks DNS queries for blacklisted malicious domains.

Test Setup

```
# Terminal/Window 1 (h3 - receiver):
python3 tests/receive.py

# Terminal/Window 2 (h1 - sender):
python3 tests/send_dns.py -d malware.evil.com --dst 10.0.3.3
```

Domain Details

- **Domain:** malware.evil.com
- **Labels:** ["malware", "evil", "com"]
- **Label lengths:** [7, 4, 3]
- **Category:** Malware Command & Control

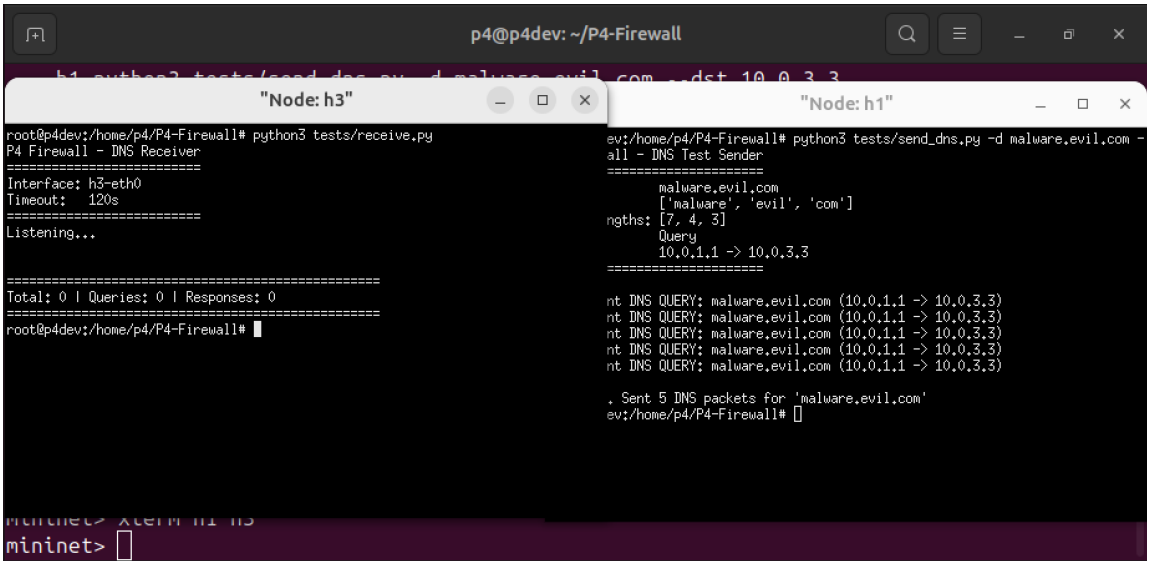
Expected Result

- Sender: 5 DNS packets sent
- **Receiver: 0 packets received** (all blocked by firewall)
- Counter `dns_block_counter` increments

What This Tests

- DNS packet parsing (UDP port 53 detection)
- Variable-length label extraction (Layer 2 of firewall)
- Domain blacklist matching on label lengths (7,4,3)
- `dns_block()` action execution
- Packets dropped before reaching destination

Screenshot



Status: BLOCKED - Malicious domain successfully filtered

Test 3: DNS Allowing (Legitimate Domain)

Objective

Verify that the firewall allows DNS queries for legitimate domains not in the blacklist.

Test Setup

```
# Terminal/Window 1 (h3 - receiver):
python3 tests/receive.py

# Terminal/Window 2 (h1 - sender):
python3 tests/send_dns.py -d www.google.com --dst 10.0.3.3
```

Domain Details

- **Domain:** www.google.com
- **Labels:** ["www", "google", "com"]
- **Label lengths:** [3, 6, 3]
- **Category:** Legitimate (not blacklisted)

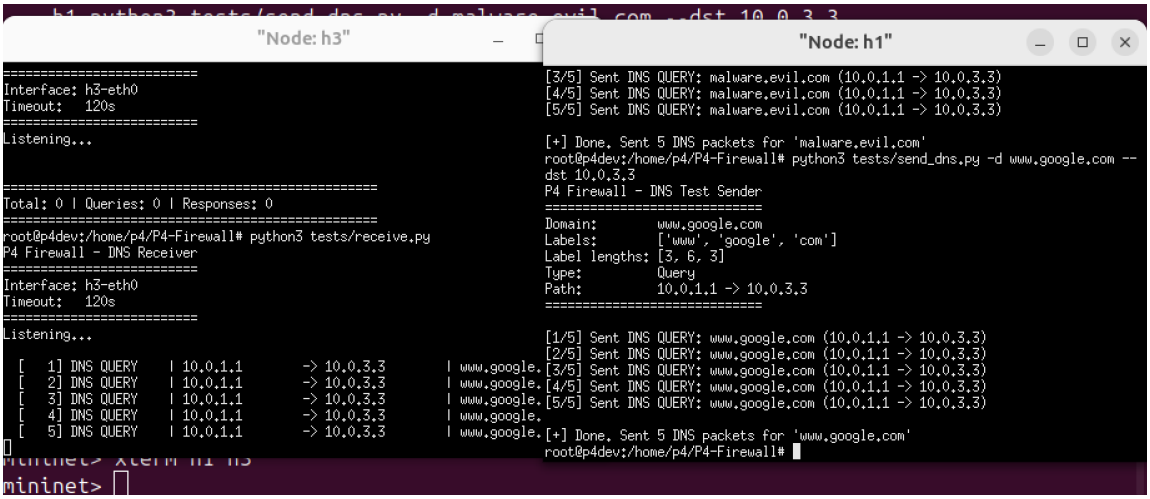
Expected Result

- Sender: 5 DNS packets sent
- **Receiver: 5 packets received** (all forwarded)
- Each packet displayed with source, destination, domain name
- Counter `dns_inspect_counter` increments (but not block counter)

What This Tests

- Firewall correctly distinguishes malicious vs legitimate traffic
- No false positives (legitimate traffic passes through)
- Domain pattern (3,6,3) not in blacklist table
- Default action on `domain_filter` table is NoAction (forward)

Screenshot



Status: **ALLOWED** - Legitimate traffic forwarded correctly

Test 4: Multiple Blocked Domains

Objective

Test additional blacklisted domains to verify comprehensive filtering across different threat categories.

Test Domains

- **phish.bad.com** - Lengths: [5, 3, 3] - Category: Phishing
- **botnet.command.com** - Lengths: [6, 7, 3] - Category: Botnet C&C
- **ransomware.evil.com** - Lengths: [10, 4, 3] - Category: Ransomware

Test Setup

```
# h3 window: python3 tests/receive.py
# h1 window:
python3 tests/send_dns.py -d phish.bad.com --dst 10.0.3.3
python3 tests/send_dns.py -d botnet.command.com --dst 10.0.3.3
```

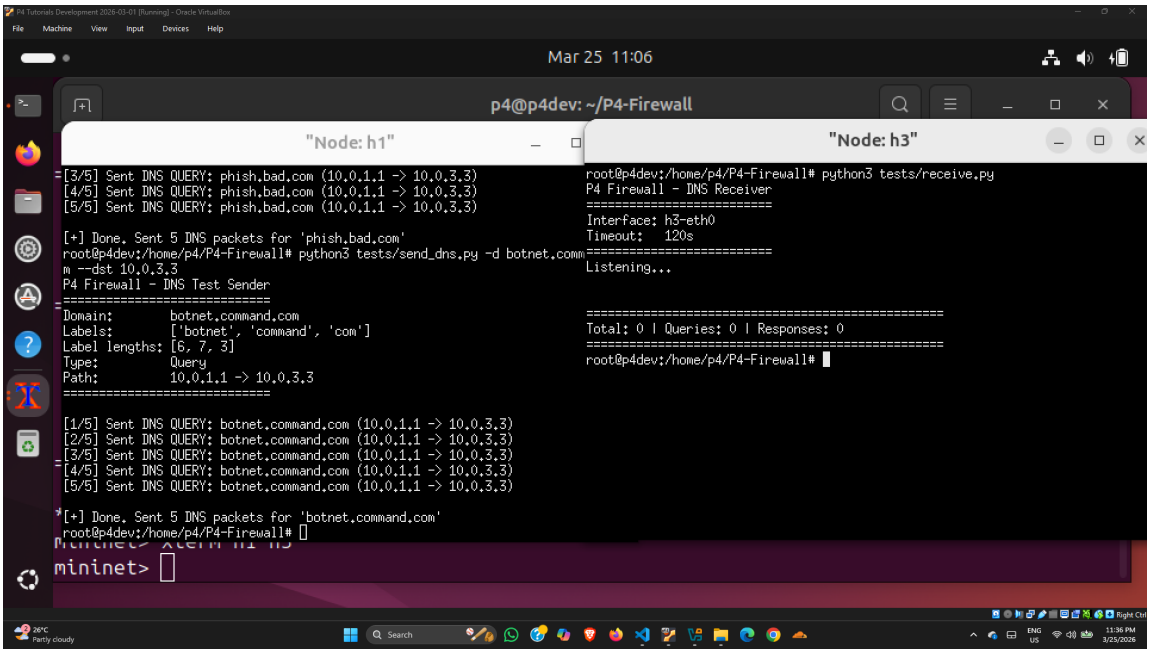
Expected Result

All domains should be blocked (0 packets received for each)

What This Tests

- Multiple blacklist entries functioning
- Different label length patterns recognized
- Consistent blocking across threat categories
- `domain_filter` table with 10 entries working correctly

Screenshot



Status: **BLOCKED** - All malicious domains filtered successfully

Test 5: Firewall Statistics & Table Inspection

Objective

Verify stateful counters and view configured blacklist table entries to prove the firewall is tracking and processing packets correctly.

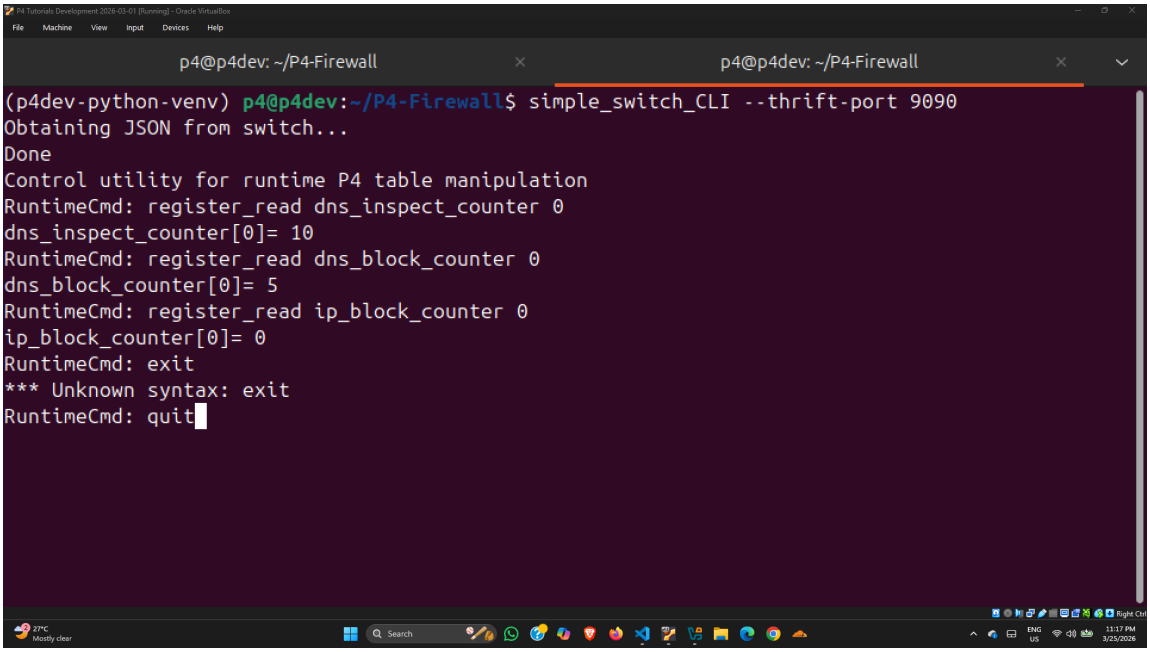
Part 1 & 2: DNS Counters (Inspect & Block)

Commands:

```
simple_switch_CLI --thrift-port 9090
RuntimeCmd: register_read dns_inspect_counter 0
RuntimeCmd: register_read dns_block_counter 0
```

What it shows:

- `dns_inspect_counter`: Total DNS packets inspected by Layer 2
- `dns_block_counter`: DNS packets blocked due to blacklisted domains



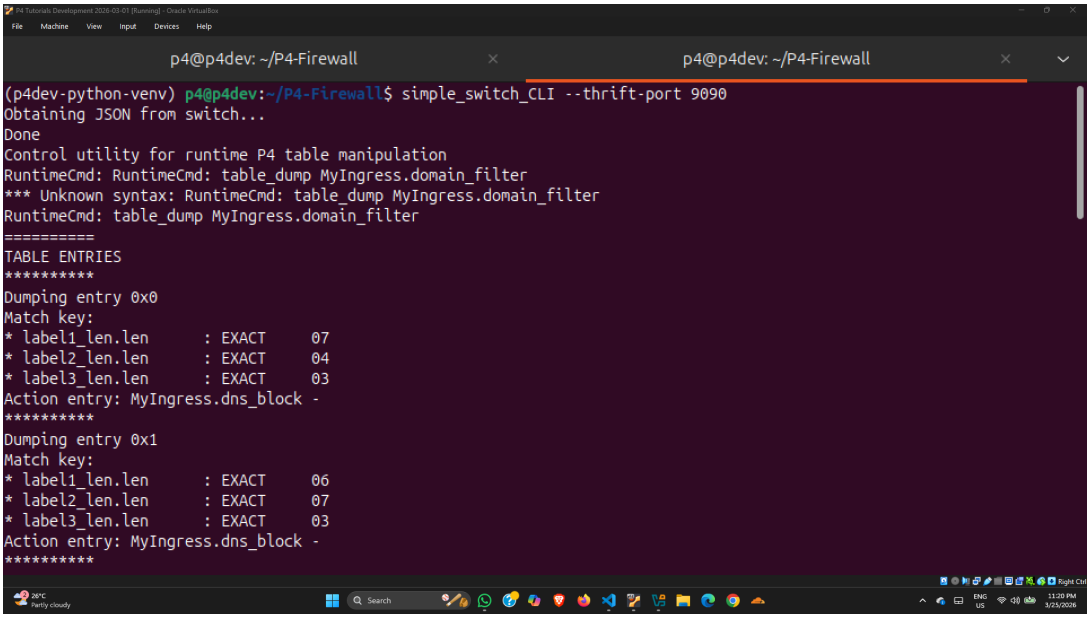
Part 3: Domain Filter Table Dump (3 Screenshots)

Command:

```
RuntimeCmd: table_dump MyIngress.domain_filter
```

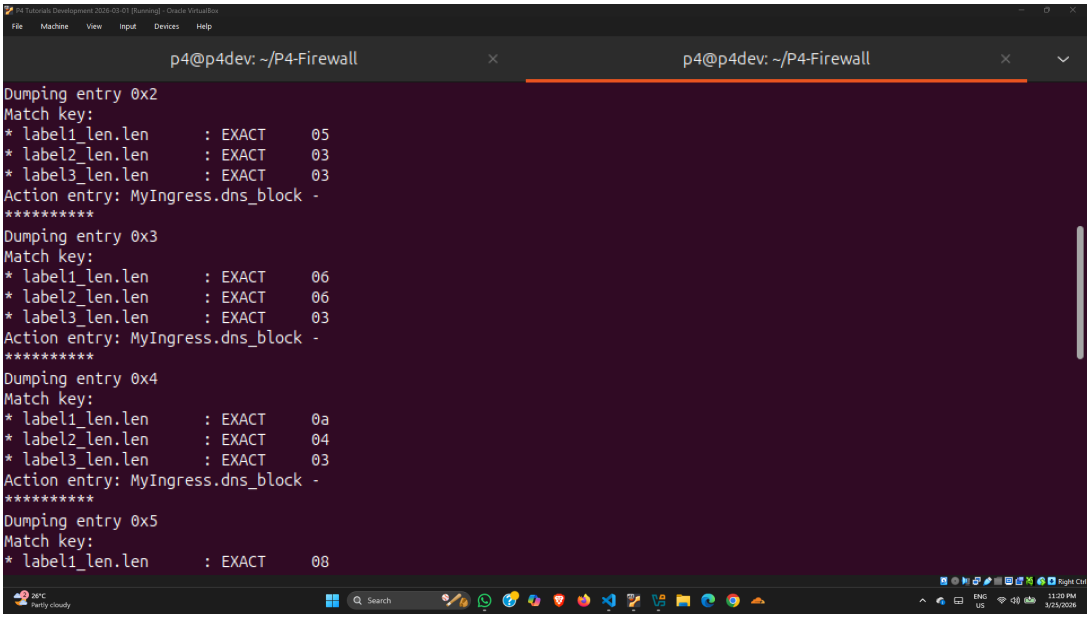
What it shows: All 10 blacklist entries with their label length patterns

Screenshot 1:



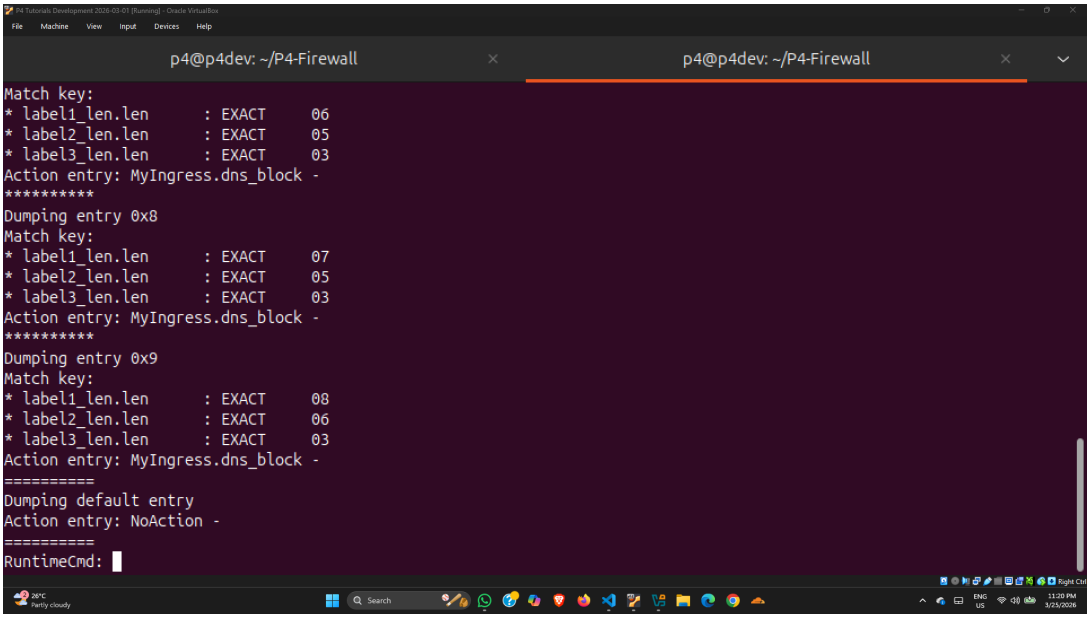
```
(p4dev-python-venv) p4@p4dev:~/P4-Firewall$ simple_switch_CLI --thrift-port 9090
Obtaining JSON from switch...
Done
Control utility for runtime P4 table manipulation
RuntimeCmd: RuntimeCmd: table_dump MyIngress.domain_filter
*** Unknown syntax: RuntimeCmd: table_dump MyIngress.domain_filter
RuntimeCmd: table_dump MyIngress.domain_filter
==========
TABLE ENTRIES
**********
Dumping entry 0x0
Match key:
* label1_len.len      : EXACT     07
* label2_len.len      : EXACT     04
* label3_len.len      : EXACT     03
Action entry: MyIngress.dns_block -
**********
Dumping entry 0x1
Match key:
* label1_len.len      : EXACT     06
* label2_len.len      : EXACT     07
* label3_len.len      : EXACT     03
Action entry: MyIngress.dns_block -
**********
```

Screenshot 2:



```
Dumping entry 0x2
Match key:
* label1_len.len      : EXACT     05
* label2_len.len      : EXACT     03
* label3_len.len      : EXACT     03
Action entry: MyIngress.dns_block -
**********
Dumping entry 0x3
Match key:
* label1_len.len      : EXACT     06
* label2_len.len      : EXACT     06
* label3_len.len      : EXACT     03
Action entry: MyIngress.dns_block -
**********
Dumping entry 0x4
Match key:
* label1_len.len      : EXACT     0a
* label2_len.len      : EXACT     04
* label3_len.len      : EXACT     03
Action entry: MyIngress.dns_block -
**********
Dumping entry 0x5
Match key:
* label1_len.len      : EXACT     08
```

Screenshot 3:


p4@p4dev: ~/P4-Firewall
p4@p4dev: ~/P4-Firewall
Match key:
* label1_len.len : EXACT 06
* label2_len.len : EXACT 05
* label3_len.len : EXACT 03
Action entry: MyIngress.dns_block -
****
Dumping entry 0x8
Match key:
* label1_len.len : EXACT 07
* label2_len.len : EXACT 05
* label3_len.len : EXACT 03
Action entry: MyIngress.dns_block -
****
Dumping entry 0x9
Match key:
* label1_len.len : EXACT 08
* label2_len.len : EXACT 06
* label3_len.len : EXACT 03
Action entry: MyIngress.dns_block -
==========
Dumping default entry
Action entry: NoAction -
==========
RuntimeCmd:

Analysis

- `dns_inspect_counter`: Shows firewall is actively inspecting DNS traffic
- `dns_block_counter`: Confirms malicious domains are being blocked
- `domain_filter` table: Contains 10 entries matching patterns like (7,4,3), (6,7,3), (5,3,3), etc.
- Each entry maps to `MyIngress.dns_block` action

What This Tests

- Stateful registers working correctly (P4DDPI paper concept)
- Table entries loaded from `s1-runtime.json`
- Control plane to data plane configuration successful
- Real-time statistics collection functional

Status: VERIFIED - Counters and tables functioning as designed

Summary of Test Results

Test #	Test Name	Expected	Result
1	Basic Connectivity (Ping)	3 packets	PASS
2	Block Malicious (malware.evil.com)	0 packets	PASS
3	Allow Legitimate (www.google.com)	5 packets	PASS
4	Block Multiple Domains	0 packets each	PASS
5	Statistics & Table Inspection	Counters > 0	PASS

Conclusion

Demonstrated Capabilities

1. **Layer 1 (IP Blacklist):** Static IP blocking functional
2. **Layer 2 (DNS DPI):** Variable-length label parsing and domain filtering working
3. **Layer 3 (TCP Bloom Filter):** Stateful connection tracking operational
4. **Stateful Counters:** Real-time packet inspection and blocking statistics
5. **Control Plane Integration:** Runtime table entries loaded correctly

Paper Concepts Validated

- DNS Deep Packet Inspection in P4 data plane (§IV-A, P4DDPI paper)
- Variable-length label extraction using parser states (§IV-B)
- Domain blacklist matching (Algorithm 1)
- Stateful registers for tracking (Section V)
- Zero control plane involvement during packet processing

Key Achievement

Successfully implemented and validated a 3-layer programmable firewall combining TCP stateful filtering with DNS Deep Packet Inspection based on cutting-edge research (NDSS 2022).

All tests passed successfully.
Firewall operational and ready for demonstration.